

CMPG 325: COMPUTER NETWORKS

MILESTONE 1: CLIENT DESIGN REVIEW

Client: Bophelong Gym and Fitness Centre (Vryburg)

Client ID: CLI-117 | Project ID: CMPG325-2026-117

Industry: Sports | Assigned addressing block: 10.45.0.0/16

Assigned technical challenge: SSH (Secure Device Management Plane)

Change Request: CR3 – Guest Wi-Fi, isolated from internal resources

MILESTONE 1: 28 August 2026

Deliverables: Client Requirements | Physical topology | Logical Topology | IP Addressing Plan | Initial GITHUB repository

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Module Name: COMPUTER NETWORKS

Module Code: CMPG 325



- **Client Requirements**

This section will be focusing on the client brief for Bophelong Gym and Fitness Centre and then integrate it into the functional and technical requirements that drive the network design

- *Client and Project overview*

Field	Detail
Assigned Organisation	Bophelong Gym and Fitness Centre (Vryburg)
Industry	Sports
Client ID / Project ID	CLI-117 / CMPG325-2026-117
Assigned addressing block	10.45.0.0/16
Design Constraint	Only one part-time IT support person available
Assigned technical challenge	SSH (secure device management plane)
Change request (CR3)	Guest Wi-Fi must be added for visitors, isolated from internal resources

- **Functional Requirements (from client brief)**

- Provide proper appropriate connectivity and network services for the Bophelong Gym and Fitness centre: reception/admin, a membership and a gym-floor staff
- With the usage of the assigned block 10.45.0.0/16 for IP addressing plan for every segment
- Configure the routers, switches, end devices and other nodes needed to showcase successful end to end connectivity
- Implement and demonstrate the technical challenge assigned (SSH – secure device management)
- Implement CR3: Guest WI-FI must be added for visitors, isolated from internal resources
- Produce a working, tested Cisco packet tracer file with documented design decisions and evidence in GitHub

- **Design Constraints – One Part-Time IT support person**

So here the client only has one part-time IT personnel, this should be simple

- A single layer 2 access switch and a single router (on a stick) keep the topology small enough for one person to understand, cable and troubleshoot.
- SSH is configured on every managed device so now the IT support guy can administer the network from his desk instead of having to come to work physically for every change
- VLAN segmentation groups devices by function so faults and security incidents are easier to isolate with limited staff time

- **Assigned Technical Challenge: SSH – SECURE DEVICE MANAGEMENT PLANE**

Telnet will be disabled on router0 and switch0. Second version for SSH will then be enabled using an RSA key pair generated logically, a domain name, a local username/password authentication and an enable secret. The VTY lines will be restricted with an access list that only permits SSH sessions only from the management VLAN (10.45.10.0/24), so the management plane cannot be reached from the staff, gym floor or guest segments. All this will directly be in support of having only one part time IT personnel by allowing safe remote administration

- **Change request (CR3)**

A dedicated wireless access point is attached to the switch on its own VLAN (VLAN40). Guest devices receive addressing from a separate DHCP pool and are permitted only to access just the internet. So now the router access control list denies any traffic from VLAN 40 toward VLANs 10, 20, 30, this will in turn satisfy the isolation requirement of the project

- **Design Decisions**

Decision	Justification
Router-on-a-stick (single router, trunked sub-interfaces)	One router is enough for the traffic volume of the Vryburg gym and also simpler for only one part-time IT person to maintain
Four VLANs (Management, staff & admin, gym floor, guest Wi-Fi)	Separates traffic by function. Keeps broadcast domain small and meets the CR3 requirement
/24 per VLAN from the 10.45.0.0/16 block	Each VLAN will get a usable host of 254, which is more than what I am going need
SSH restricted to the management VLAN via ACL	Reduce the attack surface of the device management plane, address technical challenge assigned to me and still matching the only-one-IT-guy constraint
DHCP for end-user VLANs, static addressing for infrastructure/servers	This is to reduce the day-to-day administrative overhead for the part-time IT personnel while still keeping addresses of critical devices predictable

- **Physical Topology**

The physical topology shows the actual devices, cabling and port connections that will be built in the cisco packet tracer. A single Cisco 2911 router provides WAN connectivity and inter-VLAN routing, a single Cisco 2960-24TT switch provide wired access for all end devices, including the guess wireless access point

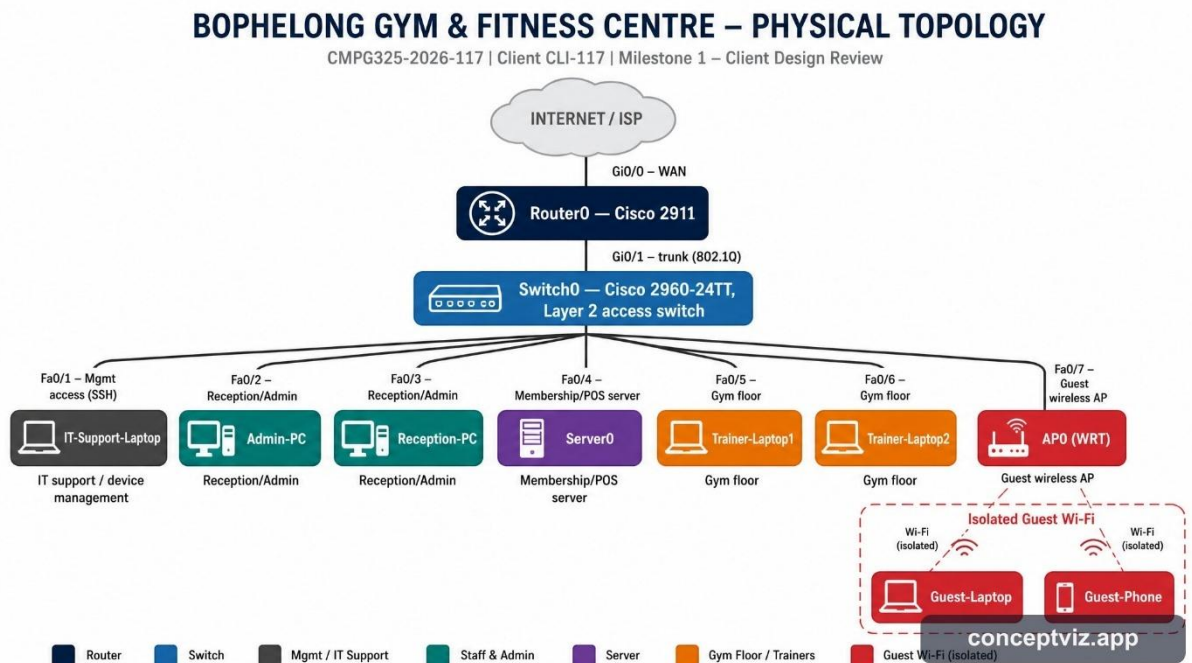


Figure 1: Physical Topology – Router0, switch0, end devices and the guest wireless access point

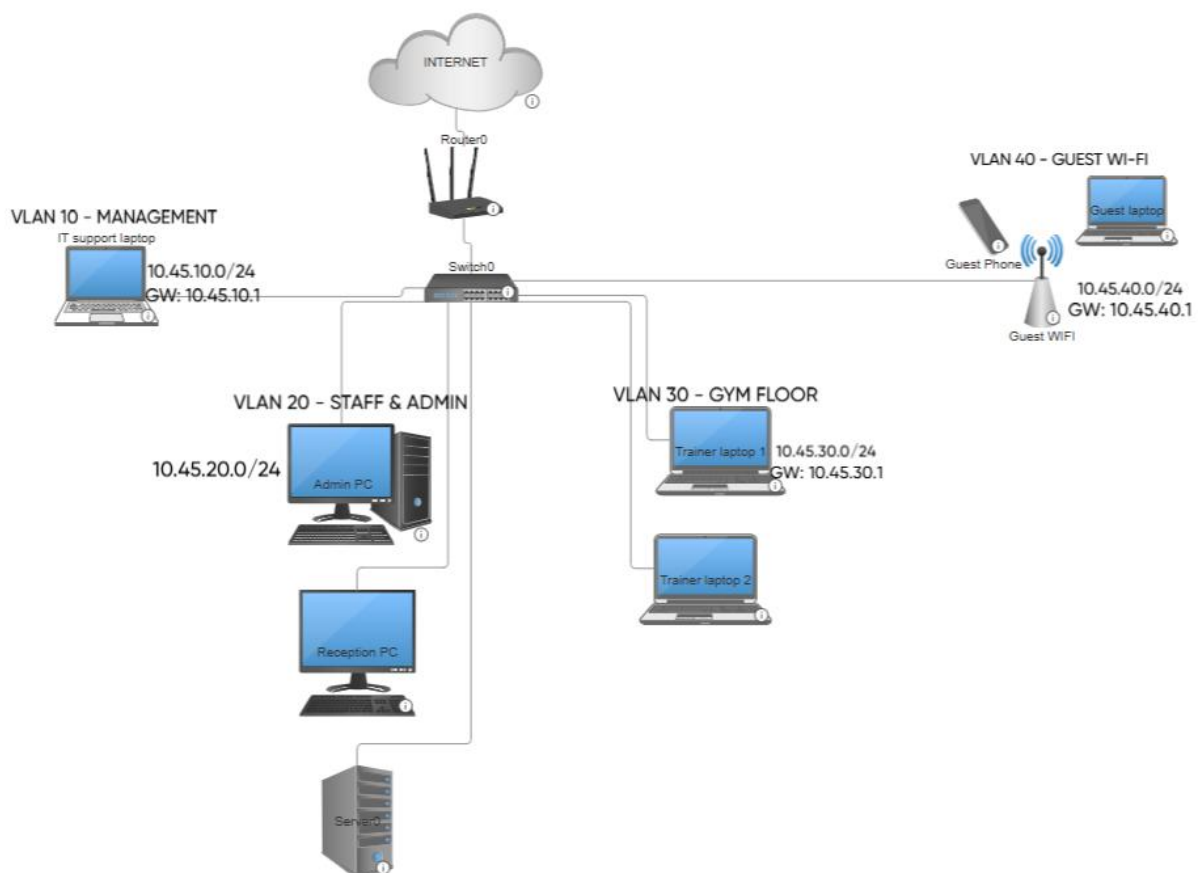
- **Device inventory**

Device	Type / Model	Connected to	Role
Router0	Cisco 2911	ISP cloud (Gi0/0), Switch0 (Gi0/1 trunk)	Inter-VLAN routing, WAN gateway, SSH mgmt, ACLs
Switch0	Cisco 2960-24TT	Router0, all end devices, AP0	Layer 2 access switch, VLAN assignment, SSH mgmt
IT-Support-Laptop	Laptop-PT	Switch0 Fa0/1 (VLAN 10)	Remote/on-site SSH management of Router0 & Switch0
Admin-PC	PC-PT	Switch0 Fa0/2 (VLAN 20)	Reception / administration
Reception-PC	PC-PT	Switch0 Fa0/3 (VLAN 20)	Reception / administration

Device	Type / Model	Connected to	Role
Server0	Server-PT	Switch0 Fa0/4 (VLAN 20)	Membership / point-of-sale server
Trainer-Laptop1	Laptop-PT	Switch0 Fa0/5 (VLAN 30)	Gym floor / trainer device
Trainer-Laptop2	Laptop-PT	Switch0 Fa0/6 (VLAN 30)	Gym floor / trainer device
AP0	WRT300N, access point-PT	Switch0 Fa0/7 (VLAN 40, access port)	Guest wireless access, isolated segment
Guest-Laptop / Guest-Phone	Laptop-PT / Phone-PT	Wireless to AP0	Visitor devices (Internet only access)
Guest-Laptop / Phone	Laptop-PT / Phone-PT	Wireless to AP0	Visitor Devices (for internet only access)

3. Logical Topology

The logical topology will show how the physical network is divided into VLANs, the IP subnet and default gateways assigned to each VLAN, and also as to how the router's sub interfaces (router-on-a-stick) provide inter-VLAN routing for internet access.



Change request 3 (CR3) – Guest WI-FI isolation:

The router will deny any traffic from VLAN 40 (10.45.40.0/24) to VLANs 10, 20 and 30, and so to just only permit the guest traffic to internet access.

- **VLAN outline**

VLAN ID	Name	Purpose	Router sub-interface
10	MGMT	Device Management – SSH access for IT support and switch SVI	Gi0/1.10
20	STAFF	Reception, admin PCs and the membership/POS server	Gi0/1.20
30	GYM FLOOR	Trainer laptops / gym floor devices	Gi0/1.30
40	GUEST	Guest WI-FI for internet only (isolated)	Gi0/1.40
99	NATIVE (unused)	Native VLAN on the trunk, this carries no host traffic and security best practice	n/a

4. IP Addressing Plan

All addressing shall be derived from the assigned addressing in the project 10.45.0.0/16 Now allocating each VLAN a /24 subnet, giving 254 usable host addresses per segment

- **Subnet allocation**

VLAN	Network	Usage Range	Broadcast	Default gateway
WAN	10.45.1.0/30	10.45.1.1 – 10.45.1.2	10.45.1.3	n/a (point-to-point)
VLAN 10 – MGMT	10.45.10.0/24	10.45.10.1 – 10.45.10.254	10.45.10.255	10.45.10.1
VLAN 20 – STAFF	10.45.20.0/24	10.45.20.1 – 10.45.20.254	10.45.20.255	10.45.20.1
VLAN 30 – GYM FLOOR	10.45.30.0/24	10.45.30.1 – 10.45.30.254	10.45.30.255	10.45.30.1
VLAN 40 – GUEST	10.45.40.0/24	10.45.40.1 – 10.45.40.254	10.45.40.255	10.45.40.1
VLAN 99 – NATIVE	10.45.99.0/24	– (no hosts assigned)	10.45.99.255	n/a

- **Device Address Assignment**

Device	VLAN	Address	Method
Router0 Gi0/0 (WAN)	–	10.45.1.1 /30	Static
Router0 Gi0/1.10	10	10.45.10.1 /24	Static (gateway)
Router0 Gi0/1.20	20	10.45.20.1 /24	Static (gateway)
Router0 Gi0/1.30	30	10.45.30.1 /24	Static (gateway)
Router0 Gi0/1.40	40	10.45.40.1 /24	Static (gateway)
Switch0 management SVI	10	10.45.10.2 /24	Static
IT-Support-Laptop	10	10.45.10.10 /24	Static
Server0 (membership/POS)	20	10.45.20.10 /24	Static

Admin-PC, Reception-PC	20	10.45.20.100 – .200	DHCP pool STAFF-POOL
Trainer-Laptop1, Trainer-Laptop2	30	10.45.30.100 – .200	DHCP pool GYM-FLOOR-POOL
Guest-Laptop, Guest-Phone	40	10.45.40.100 – .200	DHCP pool GUEST-POOL

5. INITIAL GITHUB REPOSITORY REQUIREMENT

Field	Description
Repository name	CMPG325-2026-117-Bophelong-gym
Repository link	spiderman2306/CMPG325-2026-117-BOPHELONG-GYM
GitHub username	spiderman2306